



## NATIONAL INSTITUTE OF TECHNOLOGY

KURUKSHETRA-136119

B. Tech. (PIE) – 4<sup>th</sup> Semester

END SEMESTER EXAM; AUGUST – 2026

Date of Exam: 01<sup>st</sup> August 2026

PIPC-209

Time: 180 Minutes

Production Technology - II

Max. Marks: 50

Note: Attempt all questions. Assume suitable missing data, if any.

**Q-1:** List and discuss different types of bonding materials used for manufacturing grinding wheels. Also, describe the properties of a grinding wheel. (10)

**Q-2 (a)** Explain the *Die Threading* process with its principle, construction, and working. Classify the different types of *Die Threading* methods and compare it with Thread Rolling and Thread Grinding processes in terms of accuracy, surface finish, production rate, cost, limitations, and applications. (8)

**(b)** What is meant by tooling layout in an automatic lathe? Explain the factors considered while preparing a tooling layout. Discuss tool positioning, sequence of operations, tool interference, tool travel, and production requirements. (5)

**Q-3:** A circular hole of 50 mm radius is to be punched in a copper plate of 5.6 mm thickness using a power press. The ultimate shear stress of the copper is 225 N/mm<sup>2</sup>. The desired cutting is performed at 30% penetration of the punch thickness, with normal clearance between the punch and die. (7)

- Determine the maximum punching force required when the punch has a flat face.
- If the available press capacity is limited to 45 tonnes, determine the minimum shear angle required on the punch face so that the punching operation can be performed safely within the available press capacity.

**Q-4:** Derive a suitable expression for *draw stress* in a frictionless wire drawing process. (10)

**Q-5:** A manufacturing company intends to produce a sheet metal component in large quantities. Explain the construction and working of compound and combination dies with suitable illustrated diagrams. Compare them based on the number of operations, production rate, tooling complexity, and cost. Justify the selection of the most appropriate die for high-volume production. (10)